Neuroscientific Perspectives of Emotion

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Bidirectional Projections:

- Brain Impacts on the body via visceral efferent pathways
- Body impacts on the brain through afferent feedback
  - *For Example: Laughter Yoga*
    - *Laughter is possible without humor or cognitive thought*

- Voluntary contraction of facial muscles contributes to emotional experience. (Strack et. al, 1988)

- Participants who hold a pencil with their lips, forcing their face to prevent or inhibit a smile, rate cartoons as less amusing than participants who hold pencil in their teeth, mimicking a smile (Levenson et. al, 1990).

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Embodied Cognition

- Emotion is often defined as a multicomponent response to a *significant stimulus* characterized by brain and bodily arousal and a *subjective feeling state* that elicits a tendency toward *motivated action*.
  - Body plays a crucial role

- There may be instances of emotion in which one or more are not necessary.

  - Significant stimulus (cf. emotions without cause)
  - Subjective feeling state (cf., unconscious emotions)
  - Motivated action (cf., sadness)

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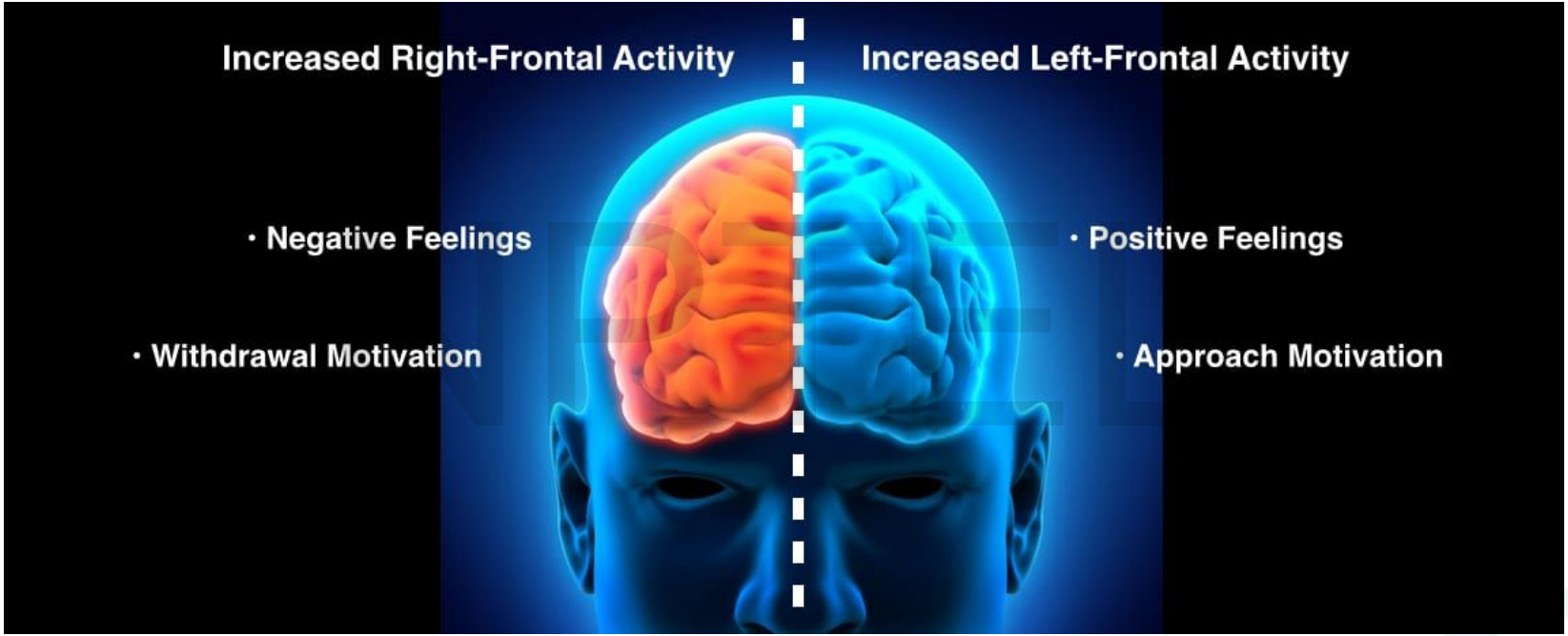
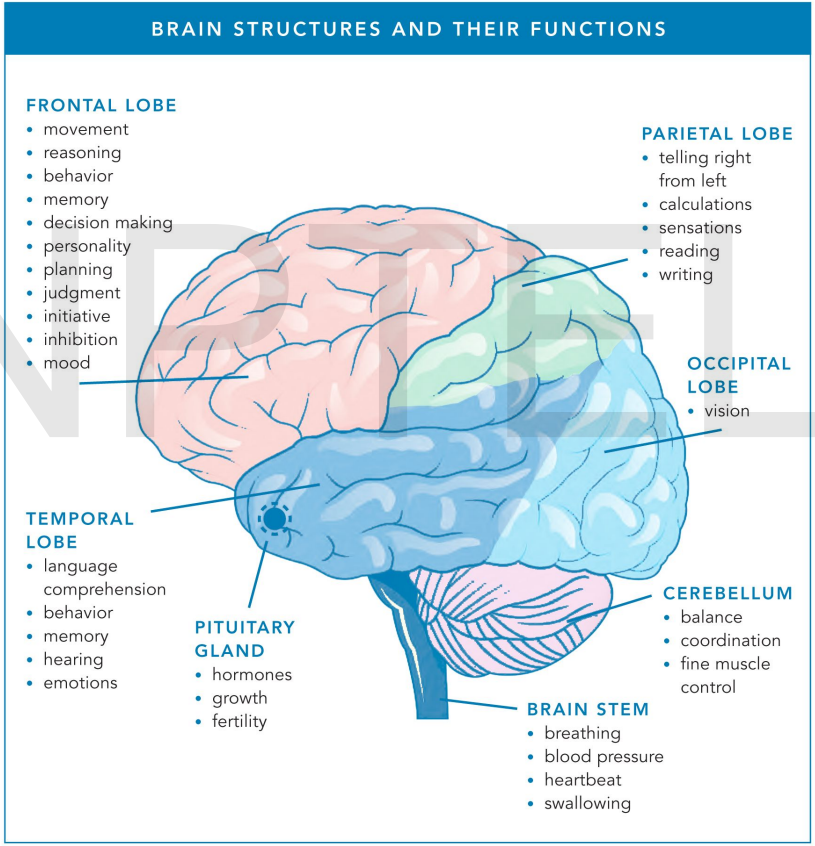
Emotion and Asymmetry



- Higher engagement of the left – relative to the right frontal brain – is related to positive feelings and higher engagement. (Coan et al., 2003).
- Exception Alert! Anger = Left Brain Bias (Harmon-Jones et. al., 1998)

The Emotional Brain



Based on an illustration from National Brain Tumor Society's *The Essential Guide to Brain Tumors*

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The Emotional Brain

Key brain regions in regards to emotion processing:

1. Prefrontal Cortex (PFC; involved in emotional experience and its regulation
2. Amygdala (Stimulus salience and motivational significance)
3. Anterior cingulate (Selection of stimuli for further processing), and
4. Insula (Feelings and consciousness)

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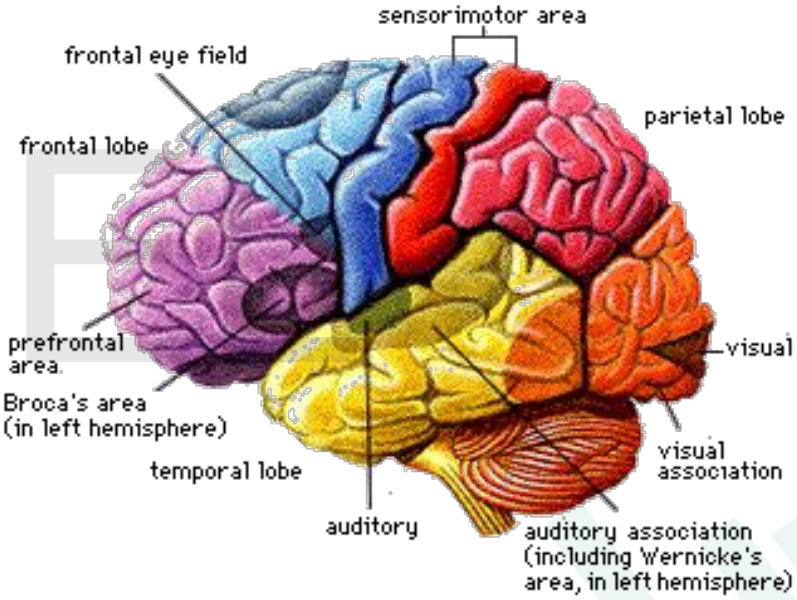
The Prefrontal Cortex (PFC)

- PFC is the most anterior part and is generally considered to play a major role in higher order cognitive activity, judgement, and planning.

- PFC is also associated in emotional experience, motivation and its regulation.



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The Prefrontal Cortex (PFC)

1. Right PFC: emotional vulnerability and affective disturbance
   a. Activity in left hemisphere region may provide a neurobiological marker of resilience (Begley and Davidson, 2012).

2. Left PFC: the emotion of anger (Harmon-Jones et al., 2010)
   a. PFC has a more important role in approach and withdrawal motivation, rather than positive and negative valence.

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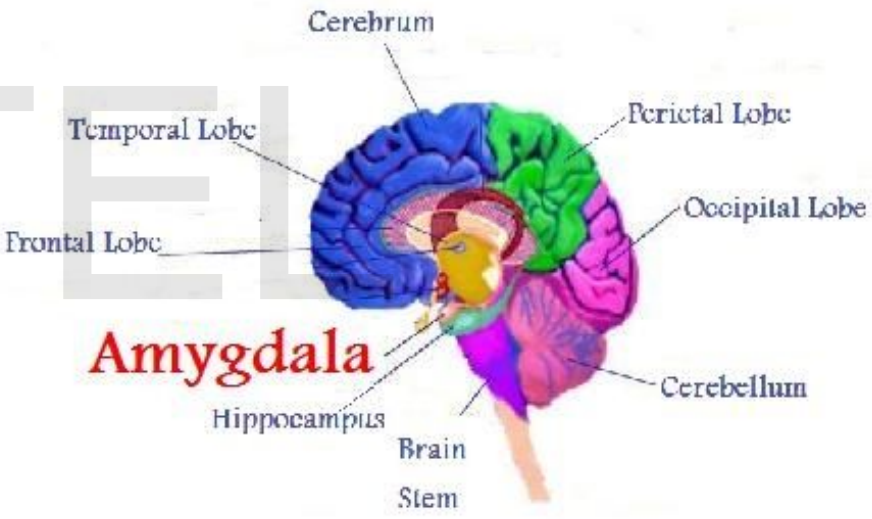
The Amygdala

- Amygdala is almond shaped cluster of nuclei located in the anterior medial temporal lobe.

- A general role in processing stimulus salience, motivational significance, and arousal (Costafreda et. al., 2008, Vytal & Hamann, 2010)



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The Amygdala

- Amygdala has a central role in negative emotions such as fear and anxiety (Murphy et. al., 2003)

- Also activated to fear, disgust, sadness, anger, happiness, humor, sexually explicit images, and social emotions. (Costafreda et. al, 2008)

- Studies have also examined amygdala activation during the experience of non-negative emotions, such as sexual arousal (Hamann et. al., 2004).

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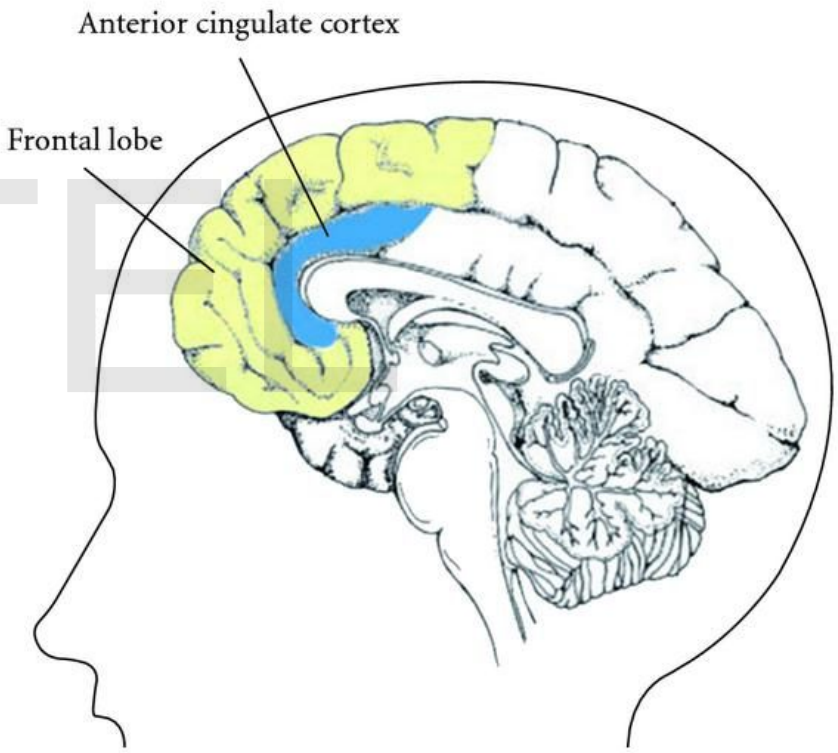
Anterior Cingulate

- AC Cortex (ACC) is key site for visceral regulations that helps to resolve which sensory input is selected for processing (Lindquist et. al., 2012).
- Activity is observed during classical fear conditioning and instructed fear based tasks and this activity is (+) correlated with sympathetic nervous system activity. (Etkin et.al., 2011)



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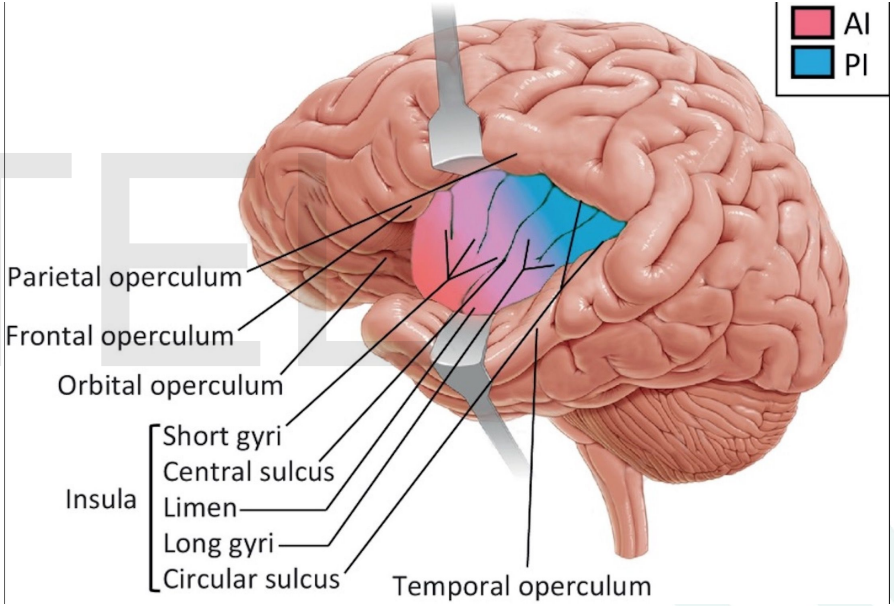
Insula

- Plays a key role in the experimental and expressive aspects of the internally generated emotion.

- The insula's role has been found in general awareness of the bodily sensations, affective feeling, and consciousness (Craig, 2009).



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Problems in traditional AC

1. Human emotions include not only the emergency emotion stimulated by intense instantaneous stimuli, but also the process emotion stimulated by the accumulation of continued weak stimuli over a period of time.

2. Emergency emotions are quick and low-precision responses. The computational complexity of traditional precision-oriented affective computing is too high to handle emergency emotions.

3. Human emotions are personalized and the emotions of different human individuals excited by the same stimulus can be different.

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Specificity of the emotions

- Although basic emotions are characterized by specific facial expressions (Ekman & Friesen, 1975), a single set of facial actions can become different emotional expressions in different contexts (Barrett, 2012): Surprise, Fear and Anger

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Fear

- Characterized by eyebrows raised and drawn together, wide open eyes, tense lowered eyelids and stretched lips. (Ekman & Friesen, 1975)

- Associated with activation within frontoparietal brain regions (Tettamanti et. al, 2012) and a broad pattern of sympathetic activation (Harrison et. al, 2013).
  - Reduced Heart Rate Variability

- Also, associated with more numerous skin conductance responses and larger electromyographic corrugator activity than is anger (Stemmler et al., 2001)

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Anger

- Characterized by lowered eyebrows drawn together, tense lowered eyelids and pressed lips. (Ekman & Friesen, 1975)

- Left frontal PFC is activated (Harmon-Jones et al., 2010)
  - Literature is contradictory (Wacker et. al, 2008)

- No change in HRV (Rainville et. al., 2011)
- Anger may elicit either an anger-mirroring or a reciprocating fear response (Harrison et al. 2013)
  - Psychophysiological responses will be different.

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Disgust

- Characterized by raised upper lip, wrinkled nose bridge and raised cheeks. (Ekman & Friesen, 1975)

- Differential skin conductance responses may depend on whether the emotion is elicited by core-disgust inducing stimuli (e.g. Pictures of dirty toilets) or body-boundary violating stimuli (e.g. mutilation scenes)
  - Unchanged or decreased skin-conductance for core-disgust inducing stimuli, e.g. dirty toilets (Harrison et. al., 2013)
  - Increased skin-conductance for body-boundary violating stimuli, e.g. mutilation scenes (Bradley et. al, 2001)

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Sadness

- Characterized by raised inner eyebrows, lowered lip corners. (Ekman & Friesen, 1975)
- Associated with increased blood flow in ventral regions (Mayberg et. al., 1999).
- Crying related sadness:
  - Increased Heart rate
  - No change in HRV
  - Increased skin conductance
- Non Crying sadness:
  - Reduction in heart rate
  - Reduced HRV
  - Reduced skin conductance
  - Increased respiration

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Happiness

- Characterized by tensed lower eyelids, raised cheeks and raised lip corners. (Ekman & Friesen, 1975)
- The intensity of smiling in photographs has been found to predict longevity (Abel and Kruger, 2010)
  - Individuals with no smiles: ranged from 72.9 years
  - Partial smiles: 75.0 years
  - Duchenne smiles: 79.9 years
- Happiness is associated with activation in medial prefrontal and temporo parietal cortices (Tettamanti et al., 2012).
- A body of work further highlights the role of the left PFC in positive affects.
- Happiness is associated with decreased HRV, amusement and joy are associated with increases (Kreibig, 2010).

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Think-Pair-Share

- For our case-study, Advance UX, which emotions can we expect to be induced and which region of brains can be active ?
- Which facial expressions can we expect to show?

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